- Proteolytic enzymes produce decomposition of proteins like meat,fish,eggs and milk.
- Amylolytic enzymes hydrolyse carbohydrates.
- Lipolytic enzymes produce a breakdown of fats into fatty acids and glycerol that are often oxidized to peroxides and aldehydes associated with rancidity or a bitter taste in fatty foods.

3. **Biological:** Biological factors that spoil food are the microorganisms like moulds, yeasts bacteria and their enzymes which settle on food at different stages. The changes brought about in the food are mainly in the form of fermentation, putrefaction and decay.



▲ Plate 8.6 Mould Growth in Bread

i. **Fermentation:** It is the anaerobic enzymatic conversion of carbohydrates into ethyl alcohol. It is caused by *Streptococcus* and *Micrococcus*.

ii. **Putrefaction:** It is the enzymatic decomposition of the proteins present in food with the production of foul smelling compounds such as hydrogen sulphide and ammonia. It is caused by *Pseudomonas* and *Proteus*.

8.2.3 Classification of foods by ease of spoilage

Foods can be categorized into three main groups on the basis of their shelf life or perishability.

1. **Stable or non-perishable foods:** Non-perishable foods include sugar, jaggery, hydrogenated fat, vegetable oil, ghee, whole grains, dhals, whole nuts and processed foods like dry salted fish / meat, papads, canned foods, jams and murabbas. These foods do not spoil unless they are handled carelessly.
2. **Semi Perishable Foods:** These foods include processed cereals, pulses and their products like flour, semolina, parched rice and popcorn. Shelf life of these products depends on the storage temperature and moisture in the air. Foods like potato, onion, nuts, frozen foods and certain canned foods can be stored for a week to a couple of months at room temperature without any undesirable changes of the products.
3. **Perishable Foods:** This includes foods such as dairy products, eggs, poultry,



▲ Plate 8.7 Spoiled Foods

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